11.12 The Ant on the Rubber Rope

WORKSHEET

The question

STEP II 2018, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/18-s2-q10

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself The string is stretched uniformly: every piece is stretched by the same factor. At time t its length is $a + ut$. If the point at distance x from the peg is a fraction $\frac{x}{a+ut}$ of the way along, how fast does it move?

STAGE 1 The ant's equation

Covers: the ant's equation

Ask yourself The ant moves at v relative to the string, plus the speed of the string beneath it. Write down $\frac{\mathrm{d}x}{\mathrm{d}t}$.

Pause and try Differentiate $\frac{x}{a+ut}$ by the quotient rule and use your equation for $\frac{\mathrm{d}x}{\mathrm{d}t}$. What is left?

STAGE 2 Reaching the end

Covers: the time to reach the end

Ask yourself Integrate from 0 to t. The ant reaches the end when the fraction $\frac{x}{a+ut}$ equals 1.

STAGE 3 The way back

Covers: the journey back

Ask yourself Walking back, the ant's speed along the string is $-v$. What is the rate of change of the fraction now? It starts at 1 at time T. When does it reach 0?

What's really going on

The harmonic series version

The puzzle is usually told with discrete stretches. A rope is 1 km long; the ant walks 1 cm in the first second; then the rope is stretched by 1 km; then the ant walks another centimetre, and so on. In the nth second, the ant covers 1 cm of a rope n km long: a fraction $\frac{1}{100\,000\,n}$. Stretching does not change its fraction, so after N seconds it has covered

$$\frac{1}{100\,000}\left(1+\frac{1}{2}+\frac{1}{3}+\cdots+\frac{1}{N}\right)\approx\frac{\ln N+0.577}{100\,000}$$

of the rope. It arrives when $H_N\approx 100\,000$, after about $e^{100\,000}$ seconds, a number with over 43 000 digits. (For comparison, the universe is about 4×10^{17} seconds old.) Our continuous answer agrees: $k=\frac{u}{v}=100\,000$, and $T\approx\frac{a}{u}e^k$.

Light in an expanding universe

The ant on the rope is a model of light crossing an expanding universe. Distant galaxies recede from us, carried by the expansion of space itself, and light from them travels towards us at speed c *through* the expanding space, as the ant walks along the stretching rope. Whether light from a distant galaxy ever reaches us depends on how the expansion rate changes with time. In a universe expanding at a steady rate, like the rope, light always arrives eventually. In a universe whose expansion accelerates, as ours appears to, there are galaxies whose light will never reach us: there is a *cosmic event horizon*. The analogue for the ant is a rope stretched exponentially, so that the end's speed grows in proportion to the length (try it in the exercises).

Gardner's ant

The puzzle seems to have been invented by the French mathematician Denys Wilquin in 1972, as a worm on an elastic band, and was popularised by Martin Gardner in his *Scientific American* column and in his book *Time Travel and Other Mathematical Bewilderments*. It is one of the best-known examples of a result that contradicts intuition and is nevertheless easy to prove, and it is a favourite illustration of the divergence of the harmonic series.

Try this next

1. **Halfway.** When is the ant halfway along the rope? Is that before or after time $\frac{T}{2}$?
2. **Exponential stretching.** Suppose instead the rope's length is $ae^{\lambda t}$. Show that the ant reaches the end only if $v > \lambda a$. (This is the cosmic event horizon.)
3. **An accelerating rope.** If the length is $a + \frac{1}{2}bt^2$, does the ant always arrive?
4. **The discrete puzzle.** In the discrete version, how many seconds does the ant need to cover 1% of the rope? Use $H_N \approx \ln N + 0.577$.

Further reading

- **Martin Gardner, *Time Travel and Other Mathematical Bewilderments*** (W. H. Freeman). Includes the ant on the rubber rope.
- **Julian Havil, *Gamma: Exploring Euler's Constant*** (Princeton University Press). The harmonic series, and the ant.
- **Barbara Ryden, *Introduction to Cosmology*** (Cambridge University Press). Expanding space, horizons and comoving coordinates, at undergraduate level.

The full guided solution is in Section 11.12:
https://beyondstep.keithmcnulty.org/ch11/rubber-rope.html